A HIGH-THROUGHPUT NANOFLUIDIC DEVICE FOR EXOSOME LOADING

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ABSTRACT

This paper presents a high-throughput nanofluidic chip named "exosome nanoporator (ENP)" enabling cargo loading into exosomes. Through the nanoconfinement of the nanochannels with a height comparable to the exosome size, transient nanopores are introduced on the exosome membrane, allowing the diffusion of cargo molecules into the exosomes from the surrounding medium. The ENP device consisting of 30,000 equivalent nanochannels demonstrates a high sample throughput. ENP-treated exosomes present enhanced drug delivery capacity to tumor cells, indicating the potential opportunities in preparing drug-carrying exosomes for biomedical applications.

KEYWORDS: Nanofluidic Device, High Throughput, Exosome, Cargo Loading

INTRODUCTION

Exosome are lipid bilayer nanoparticles naturally released by cells, serving as the intercellular transmission medium.[1] Due to their great potential in cargo delivery, exosomes have been widely used to load exogenous cargos by passive incubation and electroporation for biomedical researches. However, these methods usually suffer from the limitations of low loading efficiency and excessive damages on both exosomes and cargos. Alternatively, nanofluidics, as a technique capable of manipulating fluids and samples of interest at the nanoscale, shows great potential to prepare cargo-carrying exosomes while maintaining exosome integrity and not damaging their cargos.[2] However, due to intrinsically small feature size of nanofluidics, there is still no report on loading exogenous cargos into exosomes based on nanofluidic devices yet.

EXPERIMENTAL

The proposed high-throughput ENP is consisted of a PDMS layer and a glass substrate (Figure 1 (a)). The layout of the chip is such that a bridging nanochannel array contacts an interdigitated microchannel array (Figure 1 (b)). 20 pairs of parallel microchannel are designed on a PDMS layer, and 1500 parallel nanochannels are designed on a glass substrate with the orientation perpendicular to that of the microchannels. In such design, the ENP possesses 30,000 equivalent nanochannels, which is beneficial to improve the sample throughput.

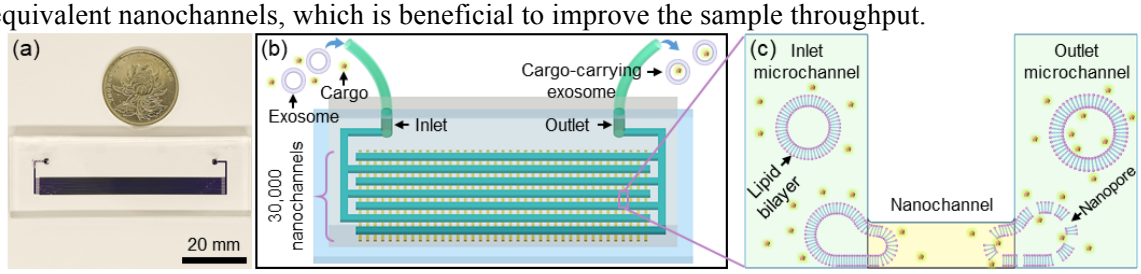

Figure 1: (a) Picture of the ENP. (b) Schematic illustrations of the cargo loading. (c) Working principle of the ENP.

The isolated exosomes from A549 tumor cells are mixed with the cargos. Then, the mixed solution is transport into ENP at a flow rate of 8 μL/min. When the exosomes pass through the nanochannels, some transient nanopores are introduced on the their membrane by mechanical compression and fluid shear, and the cargo molecules directly diffuse into the exosome through the nanopores from the surrounding medium (Figure 1 (c)). Thereafter, the cargo-carrying exosome samples (ENP group) are collected at the outlet of the chip for further applications.

In the cellular uptake experiments, the A549 cells culturing in 8-well plates are treated with 200 μL sample. As a comparison, the A549 cell are incubated with same amount of exosomes either treated by the ENP (ENP group) or passively incubated with cargo solution (incubation group) to compare the loading capacity. In addition, cells are incubated with filtered cargo solution to eliminate background noise (background group).

RESULTS AND DISCUSSION

As shown in Figure 2 (a), the exosomes display a cup-shaped membrane structure in the transmission electron microscopy (TEM) image. Furthermore, the nanoparticle tracking analysis results show a narrow size distribution of exosomes from 90 nm – 200 nm (Figure 2(b)). According to the measurements, the exosome concentration is 6.1 × 10^8 exosomes/mL, of which there are 63.7% of exosome with sizes larger than 130 nm (Figure 2(c)). Further, the cross-section of a nanochannel is characterized using a scanning electron microscopy (SEM). The nanochannel depth is approximately 130 nm, ensuring the mechanical compression on most of exosomes (Figure 2(d)).

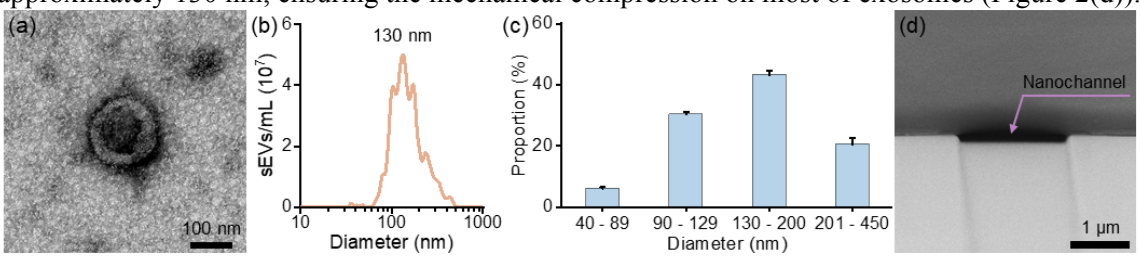

Figure 2: (a) TEM image of the sample. (b) Size distribution and concentration of exosomes. (c) Histogram on the size of exosomes. (d) SEM image on the nanochannel. Each column and error bar represents mean ± SD of three experiments

In this study, chemotherapeutic doxorubicin hydrochloride (DOX·HCl) is used as a cargo model to conduct the loading experiments. After the loading procedure, the integrity of the ENP-treated exosomes and their capability to deliver DOX·HCl into tumor cells are evaluated by cellular uptake experiments. As shown in Figure 3(a)-(c), more DOX·HCl is delivered into A549 cells by the ENP-treated exosomes. Moreover, the fluorescent intensity confirms that the amount of DOX·HCl transported into the A549 cells by the ENP-treated exosomes is increased more than 2-fold in comparison with the passive incubation method (Figure 3(d)).

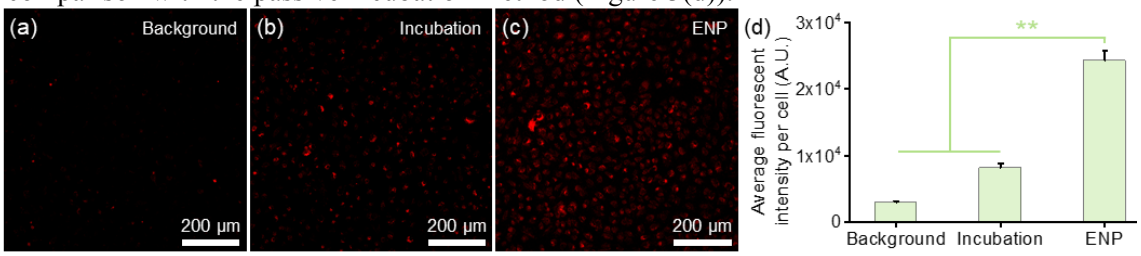

*Figure 3: (a)-(c) Fluorescent images on A549 cells obtained from confocal microscope. (d) Average fluorescent intensity per cell obtained from (a)-(c). Each column and error bar represents mean ± SD of three experiments (**p < 0.01).*

CONCLUSION

In summary, we have presented a high-throughput nanofluidic device capable of loading exogenous cargos into exosomes by permeabilizing the their membrane. Due to the chip design, 30,000 equivalent nanochannels can work simultaneously, demonstrating a flow rate of 8 μL/min for sample handling. The DOX·HCl is loaded into exosome by the ENP while maintaining exosome integrity and functionalities. Comparing to the incubation method, the ENP-treated exosome can transport more DOX·HCl into tumor cells. Hence, the proposed nanofluidic chip shows great potential of high-throughput loading various exogenous cargos into exosome for biomedical applications.

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